

Reflective Journal for Lab 04: Chihuahua or Muffin Workshop

The “Chihuahua or Muffin” workshop introduced me to the practical challenges and techniques of image classification using machine learning. The main objective of the lab was to understand how computers interpret images, how neural networks learn visual patterns, and how model architecture affects performance. By working through the notebook, modifying the model, and experimenting with different approaches, I gained hands-on experience with convolutional neural networks (CNNs), transfer learning, and the importance of data preprocessing. This workshop helped bridge the gap between theoretical concepts and real-world machine learning workflows.

One of the key techniques used in the workshop was image classification, which involves training a model to assign labels to images based on learned features. The original notebook used a simple multilayer perceptron (MLP) to classify images of Chihuahuas and blueberry muffins. Although this model demonstrated the basic idea of supervised learning, it struggled with generalization because MLPs flatten images and lose spatial information. This limitation highlighted why convolutional neural networks are the standard for image tasks. CNNs preserve spatial structure through convolutional filters, allowing them to detect edges, textures, and shapes—features that are essential when distinguishing visually similar objects like dogs and muffins.

A major concept I learned was transfer learning, which involves taking a pretrained model and adapting it to a new task. Instead of training a CNN from scratch, I replaced the original MLP with ResNet18, a deep CNN pretrained on ImageNet. This approach dramatically improved accuracy and generalization. Because ResNet18 already learned rich visual features from millions of images, I only needed to retrain the final classification layer for two classes. This saved time, reduced computational cost, and produced far better results. Transfer learning is widely used in industry because it allows high-performance models to be trained even with small datasets.

Throughout the workshop, I encountered several challenges. The biggest challenge was modifying the notebook without breaking the original structure. The teacher’s code defined transforms, dataloaders, and a model in specific cells, so inserting a new model required careful placement. At one point, I accidentally created multiple conflicting models (MLP, two CNNs, and ResNet18), which caused unexpected behavior. Another challenge was understanding why some outputs showed negative or greater-than-100% values. I eventually realized that the original model applied softmax internally, while ResNet18 outputs raw logits. Fixing this required adding softmax manually in the evaluation cells. These challenges helped me understand how model outputs work and why preprocessing and postprocessing steps matter.

One of the most valuable insights I gained is how sensitive machine learning workflows are to data transformations. The model’s performance changed significantly depending on how images were resized, normalized, and augmented. Adding data augmentation—such as random rotations, flips, and color jitter—helped the model generalize better to new images. I also learned that

validation transforms must remain consistent, and that mismatched transforms can lead to misleading results. This reinforced the idea that data preparation is just as important as the model itself.

This workshop also gave me a deeper appreciation for the real-world applications of image classification. The techniques used here are the foundation of systems like medical imaging diagnostics, facial recognition, quality control in manufacturing, wildlife monitoring, and even self-driving cars. The Chihuahua vs. Muffin problem may seem funny, but it reflects a serious challenge: teaching machines to distinguish between visually similar objects. This is directly related to problems like identifying tumors in X-rays or detecting defects in products—tasks where accuracy is critical.

Overall, this workshop was a meaningful learning experience. I not only completed the notebook but also gained confidence in modifying models, debugging issues, and understanding how deep learning systems work behind the scenes. The process of breaking the notebook, fixing it, and improving it taught me more than simply following instructions. I now have a stronger grasp of CNNs, transfer learning, and the importance of thoughtful data preprocessing. Most importantly, I continued to learn how to troubleshoot machine learning workflows—an essential skill for anyone working with AI. This lab helped me appreciate the complexity of image classification and gave me practical experience that I can apply to future projects.

References (APA Style)

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